

of the center and periphery of the capitalist world system. The differentiation between both cases, which in this research refers to the United States and Chile, is given by the external restriction of currency availability to sustain imports in the case of the capitalist periphery.

0.2 Theoretical and Conceptual Background

In heterodox macroeconomics the notion of capacity utilization is commonly derived from Keynes (2018) principle of effective demand, positing that observed output correspond to aggregate demand without necessarily fulfilling the conditions of Say's law that production creates its own demand. During Keynes landscape of the debate in economics, Say's law have been already re-interpreted by neo-classical economists through walrasian general equilibrium, basically a new version of Say's law stating that markets are capable of self-regulating excess demand or supply across markets. In the aftermath of 1929 financial crisis, Keynes gain traction as a prominent economist in the process of capitalist restructuring that would lead to the Fordist era, providing the theoretical underpinings for the wide consensus among economists of the time, that capitalist economies do not necessarily operate under full utilization of employable resources. This means that involuntary unemployment and idle capital might be persistent issues and full employment would not be achieved without government intervention. Consequently, he debunked the notion of Say's law as a law. On the contrary, Keynes was the leading economic thinker that established it as a particular case in economics in opposition to contemporary economist at the time (Roncaglia, 2006).

Marx (2000) anticipated some of Keynes's contributions on the principle of effective demand, mostly in his works presented in *Capital Volume II* where he dig deeper on his understanding of the capital circuit initially laid out in *Volume I* at the opening chapter of "the commodity". In his extension of the capital circuit, Marx developed a theoretical understanding of capitalism as a particular type of economic system featured by not only by commodity exchange, but a fiat money commodity exchange one (Basu, 2022). In this manner, he extended his initial presentation of the capital circuit in *Capital Volume I*, and explored the entangled dynamics of surplus value realization, re-capitalization, and accumulation. In identifying the trend to realization types of crisis in capitalist economies due time lags of production, distribution, and sales, as well as the need for the emergence of credit as necessary means for continued expanded reproduction of capitalist accumulation (Foley, 1982; Basu, 2014, 2019).

An interesting parallel of Marx and Keynes is that both cast doubt and theoretically broke away with their contemporaries' interpretations of Say's Law stressing of persistent disequilibrium between effective demand and productive capacities (Foley, 1985). While Marx did with the influences of Say's Law in the tradition of Classical Political Economy, Keynes did with his contemporary neoclassical economists. While it might be argued that Keynes did not completely break away with the neoclassical toolkit in *General Theory*, it was the case of his legacy embedded in the post-keynesian tradition through authors such as Nicholas Kaldor, Joan Robinson, Michael Kalecki, and Piero Sraffa.

Marx's studies on the labor process pioneered the analysis of how variations in the length and intensity of the working day at a *given level of fixed capital stock*, allowed capitalist enterprises adapting to dynamics conditions in the goods and services markets, seeking profitability while being constrained by technological and labor market conditions (Marx, 1999, 2001). In this fashion, Marx -implicitly- argued that capitalist firms had a high elasticity of capacity utilization given their capability to pump-up output in response to demand, by means of establishing the shift system and making machines work during 24 hours if necessary in order to maximize the extraction of surplus value. To do so, capitalists would be willing to incur in higher labour costs for night shifts offering wage premium to seduce wage-labourers to work during night time if necessary. Marx's argued that wage premium to particular shifts entails a utilization differential, which becomes an important factor of decision-making at the firm level, where the ratio of night-shift wages to normal wages determines the elasticity of capacity utilization by firms. Facing too high wage premium, capitalist enterprises might avoid extra shifts despite a given fixed capital capable of pumping-up productive capacities (Kurz, 1986; Nikiforos, 2013).

Capitalists' strategies to maximize profitability by means of spreading capital usage over the 24-day hours' time frame is not necessarily a homogeneous condition across sectors. On the one hand, capitalists face differential cost structure contingent to the specific form of their productive activities which are never independent of institutional and moral constraints. For example, the construction sector might raise profit rates by establishing a night shift with higher labour and input costs, providing appropriate gear to workers during the night-shift and accelerate the termination of an apartments building. However, local neighbors might organize against the night-shift to happen due noise pollution, or regulations of local city councils could forbid this from the outset without need of grassroots organization and demands to take place. Marx also considered industries with higher capital intensity face higher pressure for higher capital utilization rate, due firms need to amortize fixed costs. Nevertheless, capital-intensive sectors also face constraints due higher maintenance costs and greater risks of "moral depreciation", i.e. facing the scrutiny of public opinion due morally unacceptable working conditions for labour during particular shifts of the shift system, either by the population or the workers themselves (Trezzini and Pignalosa, 2021).

The conceptual relevance of capacity utilization is closely linked to capital utilization or output-capital ratio at effective demand level, a concept overlooked in Keynes, but central for post-keynesians. Specially for neo-kaleckian growth models as first introduced by Bhaduri and Marglin (1990), and already implicit throughout Marx's late works. While capacity utilization refers to the ratio of effective demand to productive capacities, capital utilization relates to the flow of income stream being produced at a given level of capital stock. In turn, capital utilization or the output-capital ratio, is related to choice of technique which for Marx was understood as the organic composition of capital. In contemporary Marxist Political Economy the output-capital ratio is embedded of the implicit relation between labour productivity and the rate of mechanization (Cajas Guijarro, 2024). The relation between capacity utilization and the capital-output ratio is that the latter is referred as the technological component of the profit rate, which allows to identify maximum profitability and separate it from its distributive component (Duménil and Lévy, 2003; Basu, 2018). On the other hand, capacity utilization by means of relating effective demand and productive capacities allows to "clean" the output-capital ratio from the state of effective demand and represent accurately the technological component of profitability, allowing to identify a rate of profit at normal level in distinction to the rate of profit at effective demand level¹.

Despite contemporary literature drawing on Marx's understanding of the capital circuit, reproduction schemes and his seminal work on the length of the working day and sparse capital utilization through 24 hours day implementing shift system as a devise to adjustment of capitalist firms, the standard portrait of post-keynesian understanding of Marxist political economy is fallacious and reductive by means of characterizing as a supply side theory where demand side does not matter for long-run

¹ Abstracting of price indexes and following Weisskopf (1979) profitability decomposition we can state the profit rate as:

$$r = \frac{\Pi}{K} \quad (1)$$

$$r = \frac{Y}{Y^p} \frac{Y^p}{K} \frac{\Pi}{Y} \quad (2)$$

Where $\mu = \frac{Y}{Y^p}$ corresponds to capacity utilization, $\sigma = \frac{Y^p}{K}$ to the inverse capital-output ratio or capital productivity, and $\frac{\Pi}{Y}$ to the profit share. In turn, the relation between mechanization and labour productivity is embedded in capital utilization $\rho = \frac{Y}{K}$ because:

$$\rho = \mu\sigma \quad (3)$$

$$\rho = \frac{Y}{K} \quad (4)$$

$$\rho = \frac{\frac{Y}{L}}{\frac{K}{L}} \quad (5)$$

Such that $a = \frac{Y}{L}$ correspond to labour productivity and $m = \frac{K}{L}$ to the rate of mechanization.

growth. In the own words of Blecker and Setterfield (2019, p.8) in “*Heterodox Macroeconomics*” textbook :

“In classical-Marxian analysis, meanwhile, the long run is associated with a ‘fully adjusted position’ characterized by the equalization of rates of return across sectors of the economy and equality between actual and target or ‘normal’ values of variables (such as the actual and normal rates of capacity utilization). The rate of adjustment towards such a position is not supposed to be rapid, however; indeed, the economy may never actually operate in a fully adjusted position. Finally, post-Keynesians view the long run as a historical outgrowth of sequences of short- or medium-run episodes of actual performance. They are critical of both neoclassical and classical-Marxian analyses for regarding the long run as a fixed point towards which the economy is inevitably and inexorably moving, regardless of the characteristics of its traverse path (that is, the precise route taken).”

My assessment of why post-keynesians understanding of Marxist Political Economy is conflated with Classical Political Economy has its loci in their misunderstanding of Marx’s methodological approach and lack of engagement with contemporary progress in the tradition. Despite Marx did shared the long-run method of classical authors, i.e. understanding capitalism as a system where Say’s Law hold (Foley, 2013), for Marx this was only a methodological abstraction to investigate the stability of extended reproduction in capitalist economies. Even though Marx did not solved the problem posited by himself in the reproduction schemes of *Capital Volume II* successfully, contemporary research developed in the last 50 years of this tradition has analytically proven that when considering two departments of production and consumption goods, smooth reproduction of capital accumulation is feasible (Okishio, 2022; Basu, 2022)². Nevertheless, the conditions for extended reproduction of the capitalist system in the long run, are very restrictive. For example, common features identified both in Basu (2022) and Okishio (2022) are constant rates of capitalization³ and balanced growth between both production and consumption goods sectors, among other highly unlikely features.

The question that arise from the solution of Marx’s reproduction schemes becomes then, *what is the point of portraying a smooth extended reproduction of the capitalist system in the long-run with features that are quite unlikely to hold?*. The answer can be found in Okishio (2022) who provide the proof of stability under sever restrictive conditions throughout different dynamic scenarios (with or without technological change, with balanced and unbalanced growth between production and consumption sectors), such that it is possible to identify an *Equilibrium Accumulation Trajectory* of a capitalist economy. Under this equilibrium, indeed, capacity utilization is identified as a fully adjusted position such that $Y = Y^p$, i.e. where effective demand equals productive capacities. Nevertheless, Okishio (2022) highlights that a fully adjusted position or demand equal supply equilibrium carries a contradiction in itself, grounded on the social relations under the capitalist mode of production and the development of productive forces (Okishio, 2022, Chapter 3). This is rooted that in the capitalist system production is based on a *social* division of labour, while its dynamics depend on individual private decisions where capitalist exploit wage labour not only for their private consumption, but mainly seeking profits and capital accumulation.

Let assume that the total value added (by labour) in a capitalist economy is:

$$X = C + V + S_k + S_a \quad (6)$$

²Okishio (2022) *The Theory of Accumulation: A Marxian Approach to the Dynamics of Capitalist Economy* was influential work in the tradition of Japanese Marxism being his contributions to Marxist macroeconomics based on Marx’s reproduction schemes unknown in other geographies. Only having two editions in the 70’s, being the second edition of 1976 the most widely read in Japan but published just recently in English in 2022. Some recent assessments of Okishio’s works on macroeconomics place it as a seminal contribution for heterodox macroeconomics and surplus approaches alongside Sraffa (1960) *Production of Commodities by Means of Commodities* (Asada, 2024; Kurz and Salvadori, 2024).

³The rate of capitalization refers to an index of the rate of investment over the mass of capitalists profits I/π , which is a variable seldom considered in post-keynesian macroeconomics but essential for Marxism as endogenous variable which is a proxy of funding sources for extended reproduction of the capitalist system.

Where X is total value added, C is constant capital i.e the capital stock invested by capitalists in the current period, V is the wage bill of wage-labourers, S_k is the surplus value taken from exploiting workers by capitalists in the labour process, while S_a is the profit-oriented market-mediated accumulation demand which needs to replace C to sustain simple reproduction and overcome it in order to accumulate. None of both scenarios necessarily holds because S_a is defined by private decision making, seeking profitability, but not to meet the conditions of the *Steady Accumulation Trajectory*⁴.

Hence, the general equation of value produced in a money-commodity capitalist economy is actually:

$$X > C + V + S_k \quad (7)$$

Such that S_a might sustain or change the inequality, while equating it only would be a traverse point, not a fixed one as argued by Blecker and Setterfield (2019).

Thereby, the actual trajectory of capitalist accumulation is featured by one of permanent *cumulative disequilibrium*, given by mismatches between effective demand and productive capacities. Thus, it is straight forward to assert that aggregate demand in the long-run *does matter* for Marxist Political Economy, and the long-run $Y = Y^p$ understanding of Classical Political Economy, for Marxist Political Economy it is a methodological abstraction not an understanding of what is the long-run as a time horizon. In my understanding, the core of the qualitative difference between the *Critique to Classical Political Economy* and *Classical Political Economy* is found in this substantive difference. In other words, this means to feature the long-run as one were the rate of capacity utilization traverse through, rather than as a fix point of arrival as commonly argued by post-keynesians. Hence, is absolutely misleading to assert that Marxist Political Economy is featured by a fully adjusted position in the long-run. Rather, is a system prone to booms and depressions in the short run due cumulative disequilibrium of effective demand, while an inevitable tendency to economic crisis and stagnation *in the long-run as time horizon*, which is not the same as a *methodological abstraction*.

For this characterization of an intrinsic trend towards disequilibrium as the path trajectory towards crisis and stagnation to hold it is necessary to highlight some of Okishio (2022) arguments around the following issues: i) Why effective demand follows a trajectory of cumulative disequilibrium? ii) Why economic crisis are inevitable? iii) Briefly sketch some of its triggers, and iv) Why there is an inevitability of a reversal?.

First, as aforementioned above, capitalism is based on the social division of labor but investment and production decisions are made by private capitalists seeking profits. This contradiction ensures that accumulation demand never simply equals the surplus available for investment. As a result, the gap between effective demand and productive capacities driven by increasing investments such that profits are realized, grows cumulatively. Given that capitalist plan production and investment individually on the basis of *expected profitability* for the current period, not on past or present profitability, hence with a high degree of uncertainty, capitalist enterprises trend to over-expand or over-cut during boom or depressive phases of business cycles. In a nutshell, the collective of the capitalist class behave individually as a herd behavior with tendency to overshooting their expansions/shrinkage their scales of operation and accumulation of inventories. This over(under)shooting is what defines their productive capacities, and in turn their *normal capacity utilization* despite this is by construction considered at $Y = Y^p$.

Secondly, the inevitability of crisis relies on the dependency of capitalist growth on private, profit-oriented and market-mediated decision-making on the magnitude of surplus re-investment. Here, the source of its finance through internal funds fueled by profits or credit coming of the financial system also plays a key role. The cumulative character of disequilibrium takes place because prolonged booms will eventually squeeze profits through rising costs and wages due tighten labor markets and/or

⁴The *Steady Accumulation Trajectory* is equivalent to the conceptualization of the *Golden Path* of capitalist development made by Bergmann et al. (2009) modeling of disequilibrium dynamics that to highlight the spatial unevenness of capitalist development.

rising prices of capital goods creating bottlenecks for cumulative upwards tendencies of accumulation. Likewise, a roll-back of private investment due a slowdown in the expectations of profitability might threatens the conditions of capitalist reproduction, forcing the system to "self-correction" through historical processes of crisis and capitalist re-structuring. Overall, is the class-based structure and private decision-making embedded in capitalist social relations that guarantee periodic and systemic breakdowns.

Thirdly, the triggers of capitalist crisis might be found in diverse types of historically contingent causes. For example, the overproduction in consumption-goods sectors where limited wage growth and high investment leads to a glut of inventories accumulation that cannot be realized in the market is one of them. Underproduction or bottlenecks in production-goods sectors, where rapid investments eventually runs into capacity limits is another. This is particularly important for the capitalist periphery, where capitalist enterprises might have the surplus to purchase capital goods but not necessarily have at their disposal the foreign currency to do so. Also in the periphery, infrastructure are essential for the constitution of markets in certain geographies, and the lack of it might become a bottleneck for the realization of surplus value. Finally, also we can use as an example, the most commonly known cause of excessively tight or loose labor markets. While the former might cause a tendency of rising real wages, squeezing profits and undermine capitalists power over labor putting downward pressures in profitability, loose labor markets and wide unemployment and underemployment pushes wages to a floor that threaten the material conditions of working-class reproduction. This might lead labor to revolt, or to secular stagnation of capital accumulation slow but systemic breakdowns.

Fourth, as crisis are inevitable in a certain time-horizon their reversals are as well. Crisis cannot continue indefinitely without undermining the legitimacy and functionality of the capitalist mode of production. Ultimately the same forces that causes the trigger of a crisis, such as collapsing investment or widespread unemployment, become the resolution of a previous accumulation bottleneck and decreased profitability, e.g. machinery, labor, and finance funds that once were scarce becomes available again. Alternatively, capitalist can be forced to introduce technological innovations or re-organizations of the labour process for the survival of their enterprises. One way, or another, surviving capitalists will resume their demand for accumulation seeking profitability. Hence, a cumulative downward disequilibrium produces its own causes for rebound, ensuring a subsequent recovery and cumulative upward accumulation for a new phase of business cycles ensuring that a periodic recovery is as inevitable as the crisis itself.

As can be appreciated in the following table triggers for crisis and reversals are presented some causal mechanisms of the theory of crisis and reproduction of capitalist economies by Okishio (2022). It should be noticed that their are only listed in different columns but there is no necessary correspondence between triggers causing crisis and reversals by rows. This table is presented to highlight the intrinsic but diverse nature of capitalist tendencies towards crisis and stagnation with subsequent recoveries around a notional gravitational center - which in any case under this framework is a fixed point - to reinforce the argument of cumulative disequilibrium, reversal, and inevitability of crisis in capitalist economies is reinforced and presented as a more extensive toolkit for analysis.

Table 1: Examples of triggers for crisis and reversals in Okishio (2022)

Triggers of Crisis	Triggers of Reversal
Overproduction in the consumption-goods sector	Resolution of bottlenecks (e.g., capacity, labor, credit)
Underproduction or bottlenecks in the production-goods sector	Boost in consumption-goods sector (profits stabilize)
Labor market constraints	Capitalists' private consumption (steady demand)
Low wages (threats to working-class material reproduction)	Replacement demand for worn-out fixed capital
	Technological innovations or re-organization of the labor process (boosting fresh accumulation demand)
Own elaboration based on Okishio (2022)	

In summary, is possible to assert that aggregate demand in the long-run *does matter* in Marxist Political Economy, and its centrality is located in accumulation demand which is controlled privately and decided by individual capitalists oriented by profit motive, in a commodity-exchange economy where money fulfills a function of storing value. This characteristic of the capitalist mode of production carries a contradiction with its exploitative feature, consequently, there is an intrinsic tendency to disequilibrium through cumulative paths of capacity utilization and subsequent reversals that might be explained by multiple causes given the inter-linkages between production and consumption goods sectors. For Marx, the long-run method as understood for Classical Political Economy is only a methodological abstraction. Hence, it should not be reduced simply as a time horizon, while the understanding the long-run *as time horizon* is the inevitability of stagnation and crisis (Vidal, 2014, 2019).

In the following section, we shall proceed to sketch briefly some theoretical issues and benchmark practices of literature review and the capacity of building inference from them to answer research inquiries. By doing so, I will lay-out the scope, reach and limitations of this overview on capacity utilization to assess the survey measures provided by the Federal Reserve for the case of the United States.

0.3 A brief discussion on Literature Reviews

A key distinction which is often confusing for graduate students and young researchers as myself is the precise definition of *What is a literature review?*. The answer to this question entails the scope and reach of a literature review, consequently, inform us what is the capacity of a literature review within the toolkit of researchers to nurture the research process and answer inquiries. In these lines, it is helpful to identify a distinction between a *state of the art* and *literature review*. While the former aims to provide a general landscape of the literature in a certain topic, the latter implies a more detailed analysis and a critical assessment that have been reviewed. Van Der Waladt (2021) argues that the distinction between both, particularly their aims, is fundamental to avoid confusions and have clarity about in which extent any of them can contribute to the production of new knowledge.

On the one hand, a *state of the art* is defined as an instrument to understand a wide context of a particular subject, and provide the major inputs and guidelines to formulate a research question. On the other hand, a *literature review* is described as a more complex process, aimed to critically assess the quality of methods and conclusions of previous studies. Therefore, is quite relevant that researchers are aware of what they are doing, because the standard and style of narrative between both research instruments are different. For example, a *state of the art* would engage through a descriptive approach to previous studies in a defined subject. Alternatively, a *literature review* aims to build a solid background for to justify a gap in knowledge in a particular field of subject to open the path for an theoretical or empirical study. In this fashion, the *state of the art* is a useful tool at initial stages of study in order to provide a panoramic view of the field of study. On the contrary, a *literature review* needs to critically engage the literature in order to provide a deeper understanding of the field to justify the statement of a research hypothesis and/or nurturing the building blocks of a theoretical framework. Hence, a literature review can be considered as a methodology to answer a certain scope of research questions (Van Der Waladt, 2021).

In the same fashion, Knopf (2006) also stress the criteria that a literature review should critically assess existing knowledge about a particular field of study or a particular subject in the vicinity of the research inquiry by the researcher. In the case of this literature review, it is considered as an instrument to qualitatively validate the research hypotheses previously stated. The need of developing this review is to subject to scrutiny the standard practice by post-keynesian researchers of using survey data on capacity utilization, particularly for the case of the measurements provided by the Federal Reserve of the United States. For a literature review being useful to answer research questions and contribute with new knowledge based on existing knowledge, it is necessary that its output can be assess as a critical synthesis, presenting the state of knowledge in a particular subject in organized manner such that allows to identify a potential contribution of further research.